WearTwin: A Wearable IoT-Based Real-Time Digital Twin Framework for Early Type-2 Diabetes Risk Monitoring and Alerting

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***Abstract*—Type-2 Diabetes Mellitus (T2DM) is a chronic condition that, when left undetected, can progress silently between periodic clinical visits and result in avoidable emergency complications. Existing digital-twin-based approaches to diabetes care planning, such as the DT4PCP framework, rely predominantly on retrospective Electronic Health Record (EHR) data and periodic clinical encounters to update a patient's virtual health representation, leaving a gap in continuous, real-time monitoring. Commercially available wearables that offer diabetes risk screening operate as closed, black-box systems that provide a coarse risk label without an underlying explainable or continuously evolving patient model accessible to healthcare providers. This paper proposes WearTwin, a low-cost, IoT-enabled wearable band that continuously senses heart rate, estimated blood pressure, body temperature, and activity data, and streams this data to update a real-time digital twin of the wearer's health status. The system classifies risk severity and delivers immediate, actionable alerts directly to the user's device, while simultaneously maintaining an evolving, explainable health profile that can be surfaced to healthcare providers. Unlike EHR-driven frameworks, WearTwin closes the gap between clinical visits by enabling continuous physiological monitoring at a fraction of the cost of commercial alternatives. We describe the system architecture, sensing pipeline, digital twin update mechanism, and an evaluation plan for validating the proposed framework, and we position this work as a direct extension of the research gap identified in existing EHR-based digital twin literature.**

***Index Terms*—Digital Twin, Internet of Things, Wearable Devices, Type 2 Diabetes, Real-Time Health Monitoring, Machine Learning, Early Warning System**

I. Introduction

Type-2 Diabetes Mellitus (T2DM) is among the most common chronic diseases in the world, which, if improperly managed, may result in serious complications such as cardiovascular diseases, kidney failure, and neuropathy, often requiring assistance in emergency departments [1]. Classic treatment of diabetes is based on regular visits to physicians and standard guidelines that are always retroactive: meanwhile, in the time between visits, a patient's condition may worsen.

The emergence of DT technology in healthcare has enabled a shift from reactive to proactive chronic disease management. A digital twin is a continuously updated virtual representation of a physical entity in this context, a patient that integrates historical and real-time data to simulate, predict, and support clinical decision-making [1]. Recent work, such as the DT4PCP framework proposed by Alizadeh et al. [1], has demonstrated that digital twins constructed from retrospective EHR data, social determinants of health (SDoH), and periodic clinical encounters can predict near-future ED visit risk for T2D patients with strong accuracy (AUC of 0.82 using ensemble and gradient-boosted models). However, this framework – like most EHR-driven digital twin systems – updates the patient's twin only when new clinical or hospital data becomes available, leaving the interval between visits effectively unmonitored. The authors of DT4PCP explicitly identify the incorporation of real-time wearable device data as an open direction for future work.

In parallel, commercial wearable manufacturers have begun introducing on-device diabetes risk screening. Huawei's Watch GT 6 Pro, for instance, uses photoplethysmography (PPG) sensor data collected continuously over 3–14 days to produce a Low/Medium/High risk classification [4]. While effective as a consumer screening tool, such systems function as closed, proprietary black boxes: they do not expose an explainable, continuously evolving patient model to healthcare providers, and they are not designed to interoperate with a clinical digital twin framework.

This paper addresses the space between these two paradigms. We propose **WearTwin**, a lowcost, open, IoT-based wearable framework that continuously senses physiological signals and uses them to update a realtime digital twin of the wearer, while issuing immediate, human-readable alerts to the user. The specific objectives of this work are:

1) To design and prototype a low-cost wrist-worn IoT device capable of continuously sensing heart rate, estimated blood pressure, body temperature, and motion/activity.
2) To design a digital twin update mechanism that ingests this streaming sensor data in place of, or in addition to, periodic EHR updates.
3) To implement a risk classification and alerting pipeline that notifies the wearer in real time with actionable, severity-graded guidance.
4) To outline a pathway by which the resulting digital twin can be surfaced to healthcare providers for early, proactive intervention.

II. Literature Review

A. Digital Twin Frameworks in Healthcare

The technology of Digital Twins has been seen as more and more feasible for the organization of personalized healthcare initiatives [2]. Vallée [3] mentions the larger context and practicality of using digital twins in the personalized medicine domain, despite the fact that contemporary uses of this technology still rely heavily on retrospective, encounter-specific data instead of continuous data streams. Alizadeh et al. [1] introduce DT4PCP, the common digital twin system for personalized care planning, and show that its implementation in T2D management, using the EHR dataset obtained from the regional health information exchange. Their model allows connecting retrospective data of clinical, demographic, and social determinants of health with periodically updated data to predict risks of visiting emergency departments, showing AUC equal to 0.82 and employing ensemble learning, random forests, and XGBoost models in research. By analyzing the importance of different variables for the study, authors have highlighted the importance of vital signs, healthcare utilization, socioeconomic indicators, and the like.

B. Wearable-Based Diabetes Risk Screening

Aside from the academic research on digital twins, manufacturers of consumer level devices have started to implement the capability of diabetes risk estimation in their devices. For example, the diabetes risk estimation tool by Huawei presented at the World Health Expo 2026 uses a combination of continuous PPG sensing over the time period of 3–14 days that allows it to classify the diabetes risk as low, medium, or high [4]. This technology is based on scientifically demonstrated connection between raised resting heart rate, decreased heart rate variability, and the impaired recovery of activities as digital markers of metabolic malfunction. Similar technology is said to be developed by organizations such as Garmin, Samsung, and Google though none of the technology has reached commercial deployment with comparable clinical validation of it.

C. Wearable Health Monitoring Systems with the use of IoT

Many research initiatives reported the use of low-cost IoT devices for the continuous tracking of biological parameters. For instance, Armand et al. [9] created a low-cost, wireless system for monitoring cardiovascular activity, which consists of wearable sensors connected to a mobile application. The system is well suited for use in Cameroon. One of the advantages of the system is tracking of systolic, diastolic, and heart rate values as well as informing medical personnel of any abnormalities. On the other hand, Allbadi et al. [10] proposed a wearable ECG and heart rate monitoring device based on the ESP32 microcontroller connected to the Arduino IoT Cloud that enabled remote collection of measuring signals from patients by medical doctors.

D. Machine Learning and Explainable AI for Diabetes Risk Prediction

Gradient-boosting tree models have shown consistent success on even well-structured clinical and physiological datasets. In their study, Rafie et al. [11] used XGBoost and CatBoost algorithms in conjunction with class balancing using SMOTE to predict the risk of Type 2 diabetes in a Middle Eastern cohort dataset. This method produced very accurate results and employed SHAP (SHapley Additive exPlanations) to make the information obtained from this prediction accessible to end users, which significantly affirms the use of XGBoost models used in this research as preliminary ones for risk classification and illustrates the role of explaining the results obtained as a necessity instead of an extra advantage because of the targeted users of the product who are patients and health care providers who need to understand how their labels have been calculated.

III. Research Gap

Based on the reviewed literature, three specific gaps motivate this work:

1) **Update latency.** EHR-driven digital twin frameworks such as DT4PCP update the patient's virtual representation only when new clinical data is recorded, typically during a hospital or clinic encounter. This leaves the interval between visits effectively invisible to the predictive model.
2) **Lack of continuous, low-cost, open monitoring.** Commercial wearables that provide continuous risk screening are proprietary, closed-source, and priced beyond the reach of many patients and healthcare systems, particularly in resource-constrained settings.
3) **Disconnection from a clinician-facing digital twin.** Existing consumer wearables surface a coarse risk label to the end user but do not maintain or expose an explainable, evolving health profile that a healthcare provider can inspect, trend, or act upon – a capability that clinically-oriented digital twin frameworks do provide, but only from periodic data.

WearTwin is proposed to directly address these gaps by combining continuous, low-cost wearable sensing with a digital-twin update pipeline inspired by, and designed to extend, the DT4PCP framework – explicitly fulfilling the future work direction identified by Alizadeh et al. [1] of incorporating real-time wearable device data.

Table I summarizes how WearTwin is positioned relative to the two closest reference systems discussed above.

TABLE I
COMPARISON OF WEARTWIN WITH EXISTING REFERENCE SYSTEMS

Aspect	**DT4PCP [1]**	**Huawei GT6 Pro [4]**	**WearTwin (proposed)**
Data source	Retrospective EHR + periodic visits	Continuous on-device PPG	Continuous wearable multi-sensor
Update frequency	Per clinical encounter	Rolling 3–14 day window	Real-time (per sensing cycle)
Output	Risk score + SHAP-ranked features	Low/Med/High label	Risk label + trend-based digital twin
Provider access	Yes (GUI dashboard)	No	Yes (proposed dashboard)
Openness	Research prototype	Closed, proprietary	Open, low-cost, reproducible
Approx. cost	N/A (software only)	High (flagship watch)	˜INR 2,500 (hardware)

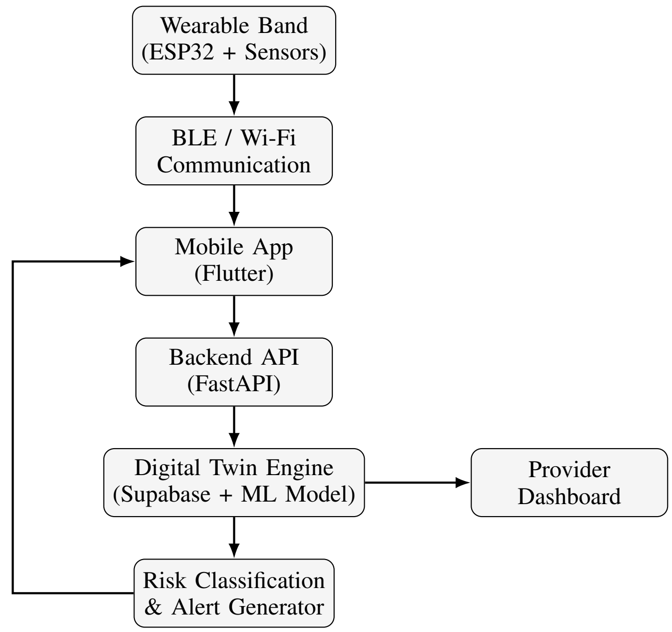

Fig. 1. Proposed WearTwin system architecture: sensor data flows from the wearable band through the mobile application to the backend, where it continuously updates the digital twin, drives real-time alerts back to the user, and exposes a provider-facing view.

IV. PROPOSED SYSTEM AND METHODOLOGY

A. System Overview

Fig. 1 illustrates the proposed WearTwin architecture, comprising four layers: (1) the wearable sensing layer, (2) the communication and mobile application layer, (3) the backend digital twin and inference layer, and (4) the provider-facing dashboard layer.

B. Hardware Design

A wearable prototype comprising of an ESP32 microcontroller (Wi-Fi and BLE compatibility) is designed using a MAX30102 sensor to measure heart rate and SpO_2 , a MLX90614 infrared sensor for temperature, an MPU6050 for motion detection, and HW-887 for indirect blood pressure measurement. To power up the system, a LiPo battery of 3.7V and 500mAh is installed along with a TP4056 charging unit and a body designed using 3D PLA printing technology. Firmware of the prototype uses different strategies to save power and only activate sensors at pre-defined intervals and remain in hibernation until the companion application connects.

C. Data Flow and Digital Twin Update

Sensor readings are packaged as a JSON payload and transmitted via BLE to the companion mobile application, which forwards the data to a backend REST API. Unlike the DT4PCP framework, where the digital twin is updated upon clinical encounter, WearTwin updates the twin on every sensor transmission cycle. The digital twin record maintains: (i) the latest raw vitals snapshot, (ii) rolling short-term averages (e.g., 24-hour, 7-day), (iii) a time-stamped history of risk scores, and (iv) anomaly flags for sudden deviations (e.g., a sustained resting heart-rate spike).

D. Risk Classification

Incoming physiological features – heart rate, estimated blood pressure, body temperature, and activity level, combined with static profile data such as age, BMI, and family history – are passed to a trained classification model to produce a risk label (Low / Moderate / High) and an associated confidence score, following the same feature categories established as predictive in [1]. Candidate algorithms considered include XGBoost, LightGBM, and logistic regression as a baseline, consistent with tree-based ensemble methods shown to perform well for structured physiological and clinical data.

Algorithm **??** summarizes the end-to-end risk classification and alert-triggering logic executed on the backend for each incoming sensor transmission.

E. Alerting Mechanism

When a risk transition is detected (e.g., Low $\rightarrow$ Moderate), the system generates a notification delivered directly to the user's wearable/device, containing: (i) the detected condition in plain language, (ii) a recommended immediate action, and (iii) an indication of potential severity if the condition is not addressed. This closes the loop between passive sensing and active, real-time patient guidance – the key capability absent from EHR-triggered digital twin updates.

V. EVALUATION PLAN AND EXPECTED OUTCOMES

As this work presents a system design and initial prototype rather than a completed clinical deployment, we outline an evaluation plan rather than reporting outcomes from live patient trials.

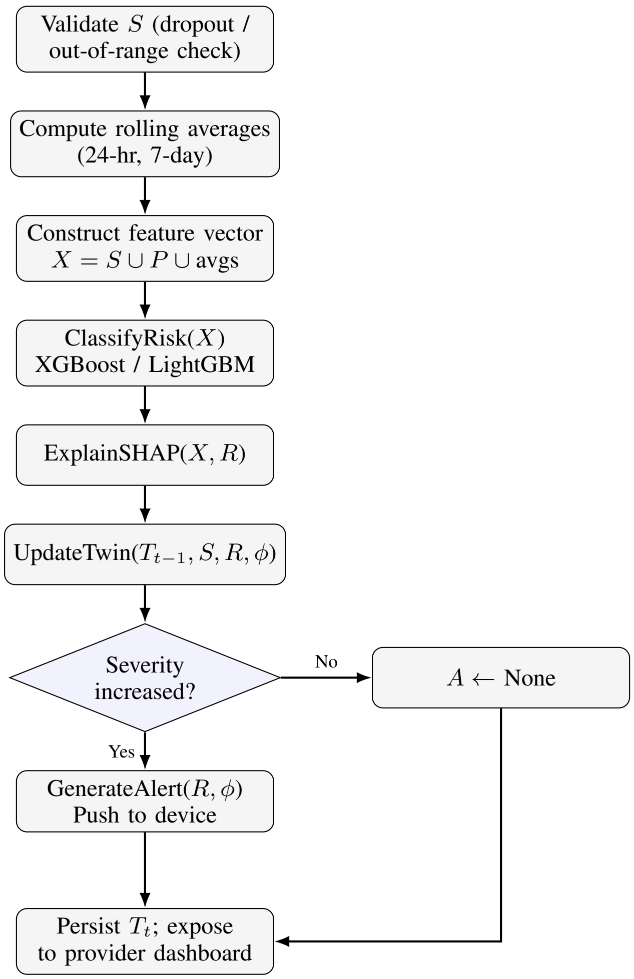

Fig. 2. WearTwin risk classification and alerting logic.

A. Bench and Functional Validation

The first step will be testing only on simple functional correctness (checking that there is stable BLE communications between the sensor and mobile app, ensuring correct data transfer from sensor to digital twin, and that alerts are triggered correctly in simulated threshold situations).

B. Sensor Accuracy Benchmarking

All sensor readings will be compared to results from regular consumer-use devices (like a clinical thermometer, oscillometric BP cuff). When a systematic bias can be seen, the necessary correction value will be calculated.

C. Expected Outcomes

Based on the architecture and the precedent set by comparable systems in the literature, we expect: (i) the continuous update cycle to substantially reduce the latency between a physiological change and its reflection in the digital twin, relative to encounter-triggered EHR updates; (ii) the low-cost hardware bill of materials (approximately INR 2,500 per unit) to demonstrate that continuous digital-twin-driven monitoring is feasible outside high-cost commercial wearable ecosystems; and (iii) the explainable, provider-facing twin view to offer a transparency advantage over closed commercial risk-screening products.

We explicitly do not claim diagnostic accuracy or clinical validation at this stage; these require a larger-scale, ethics-board-approved clinical study, which is identified as future work in Section VII.

VI. Discussion

The WearTwin structure would arrive with many benefits as opposed to the methods that are already used today. Compared to EHR-driven digital twin frameworks such as DT4PCP, it removes the dependency on clinical encounters to refresh the patient model, enabling continuous monitoring during the intervals when patients are most likely to be unobserved. Compared to commercial wearable risk-screening products, it is substantially lower-cost, open in its methodology, and designed from the outset to expose an explainable, trend-based patient model to healthcare providers rather than a single opaque risk label.

Several limitations must be acknowledged. Initially, the sensors that are utilized, in particular, the indirect PPG-based estimator of blood pressure, are specifically designed to provide estimates and not clinical readings. They should be calibrated against standard equipment. Moreover, the readings must be regarded as indicating trends. Additionally, the system has not yet been clinically tested on a patient population; any helpful application would require regulatory scrutiny. The prototype described here is clarified as a research tool and proof of concept for educational purposes, rather than a certified medical instrument, in accordance with India's Medical Device Rules (CDSCO) that control the devices making claims about diagnosis. Third, battery life under active BLE transmission remains a constraint (approximately 2 hours continuous, extendable to 15–16 hours with duty-cycled deep sleep), which will influence real-world wearability. Finally, the machine learning model underlying the risk classifier has not yet been trained on data collected from the physical prototype itself; initial training is expected to rely on public diabetes datasets (e.g., the PIMA Indians Diabetes Dataset [12]), with fine-tuning on real sensor data planned as a subsequent phase.

In addition to the constraints imposed by the algorithms, the implementation of an explainable classification pipeline based on SHAP is a design solution to the well-known problem of gradient-boosted models in the medical field: although gradient-boosted models demonstrate outstanding predictive performance, they lack ascertainable interpretation of clinical outcomes, which makes them unappealing for medical professionals [11]. By offering the top features in every risk transition, WearTwin maintains trust in the reasoning that the black box consumer risk label produced by commercial devices lacks.

It is also worth situating the hardware design choices against the broader body of low-cost IoT health monitoring literature. Prior ESP32-based systems have demonstrated that continuous cardiovascular monitoring is achievable at low cost even in resource-constrained deployment settings [9], and that BLE/Wi-Fi-based wearable ECG and heart-rate nodes can reliably bridge patient and physician over standard IoT cloud platforms [10]. These precedents support the feasibility of the proposed architecture, though it should be noted that none of the cited systems specifically target a continuously updated, explainable digital twin representation for T2D risk – which remains the specific contribution of this work.

VII. Conclusion and Future Work

This paper proposed WearTwin, a low-cost, wearable IoT framework designed to extend existing EHR-driven digital twin approaches to Type-2 Diabetes management – specifically the DT4PCP framework – into the domain of continuous, real-time patient monitoring. By streaming physiological data from a wrist-worn device directly into an evolving digital twin, the proposed system aims to close the monitoring gap between clinical encounters while remaining substantially lower-cost and more transparent than existing commercial wearable diabetes-screening products.

Future work includes: (i) completing hardware assembly and firmware development for a fully functional prototype; (ii) training and validating the risk classification model on both public datasets and data collected from volunteer testing; (iii) conducting sensor calibration studies against clinical-grade reference instruments; (iv) developing the provider-facing dashboard to enable clinician review of the digital twin; and (v) pursuing a larger-scale, ethics-approved clinical validation study, consistent with the direction identified as future work by Alizadeh et al. [1].

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